Winning Space Race with Data Science

Data science capstone project

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Executive Summary

Historical launch variables contain predictive patterns associated with landing outcomes

Methodologies

Collected and analyzed SpaceX Falcon 9 launch data using APIs, web scraping, SQL, and Python.

Performed exploratory data analysis to identify factors influencing first stage landing success.

Built interactive visualizations using Folium maps and Plotly Dash dashboards to explore launch patterns and operational trends.

Developed classification models to predict Falcon 9 first stage landing outcomes using machine learning techniques.

Results

Logistic Regression and K-Nearest Neighbors achieved the highest prediction accuracies, suggesting that historical launch characteristics contain predictive patterns.

Introduction

Reusable rockets reduce launch costs considerably

Background and context

SpaceX significantly reduced launch costs by developing reusable Falcon 9 first stages capable of landing and being reused across future missions.

Predicting landing success is therefore operationally and economically important, as unsuccessful landings increase mission costs and reduce reusability efficiency.

Research Objectives

This project analyzes historical Falcon 9 launch data to identify the factors associated with successful landings and develop machine learning models capable of predicting landing outcomes based on launch characteristics.

Section 1

Methodology

Methodology Overview

Analytical workflow applied to investigate Falcon 9 landing outcomes

DATA COLLECTION

Launch records were collected from the SpaceX REST API and complemented with additional mission information extracted through web scraping.

DATA WRANGLING

Missing values, inconsistent records, and categorical variables were cleaned and transformed into structured features for analysis.

EXPLORATORY DATA ANALYSIS

SQL queries, statistical analysis, and visualizations were used to identify relationships between launch conditions and landing outcomes.

PREDICTIVE ANALYSIS

Classification models were trained and compared to predict Falcon 9 landing success using historical launch variables.

Data Collection

The project combined multiple data collection techniques to build a structured launch dataset

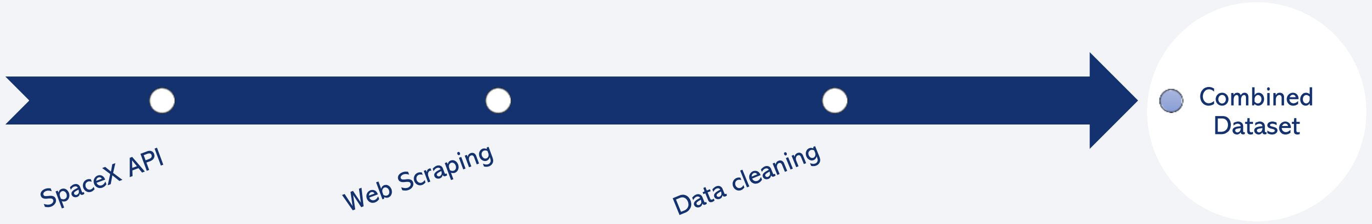
Data collection sources

Launch data was collected from the SpaceX REST API and additional mission information was extracted through web scraping techniques.

The collected information included launch dates, booster versions, payload mass, launch sites, orbit types, and landing outcomes.

Python libraries including *Pandas*, *Requests*, *BeautifulSoup*, and *SQL* tools were used to retrieve, clean, and organize the data into structured datasets for further analysis and machine learning modeling.

Data collection process



Data Collection – SpaceX API

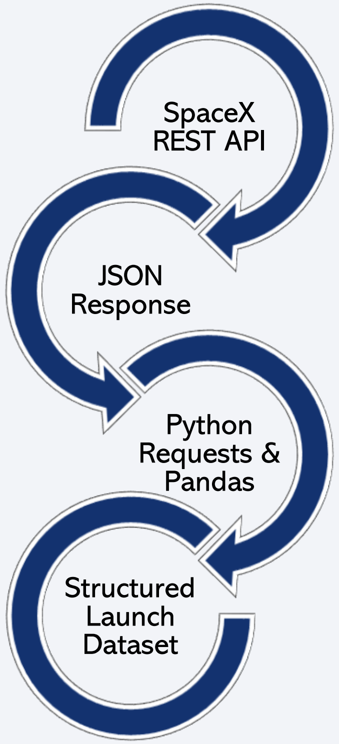
Launch records were retrieved programmatically from the SpaceX REST API

Launch data was retrieved using the SpaceX REST API and Python requests.

API responses provided structured launch information in JSON format.

Relevant variables such as launch date, booster version, payload mass, orbit type, and landing outcome were extracted and organized into Pandas DataFrames.

The processed dataset was exported for exploratory analysis and machine learning modeling.



GitHub ⬅ *Click here to view complete API extraction notebook on GitHub*

Data Collection – Web Scraping

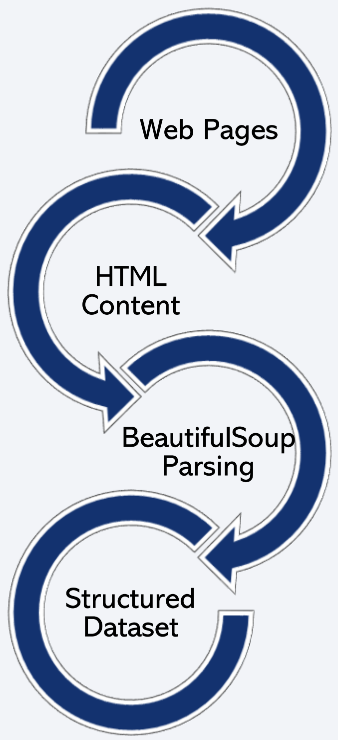
Additional launch information was extracted from publicly available web pages using web scraping techniques

Web scraping techniques were used to collect additional Falcon 9 launch information from publicly available web pages.

HTML content was parsed using BeautifulSoup to extract relevant launch variables and historical records.

Extracted data was cleaned, structured, and combined with API data for further analysis and visualization.

The final dataset was prepared for exploratory analysis and machine learning modeling.



GitHub ⬅ *Click here to view complete web scraping notebook on GitHub*

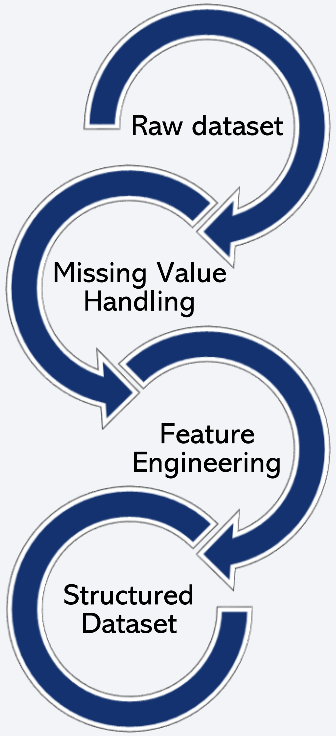
Data Wrangling

Raw launch data was cleaned and transformed into structured datasets for analysis and machine learning

Missing values and inconsistent records were identified and processed during data preparation.

Relevant launch variables were cleaned, standardized, and transformed into structured features.

A binary landing outcome variable was created to support classification modeling.



GitHub ⬅ *Click here to view complete data wrangling notebook on GitHub*

EDA with Data Visualization

Visual exploration techniques were used to identify patterns influencing Falcon 9 landing success

Scatter plots, bar charts, line charts, and pie charts were used to explore relationships between launch variables and landing outcomes.

Visual analysis identified patterns associated with payload mass, orbit type, launch site, and mission chronology.

The exploratory analysis supported feature selection and preparation for predictive modeling.

GitHub *Click here to view complete eda with data viz notebook on GitHub*

EDA with SQL

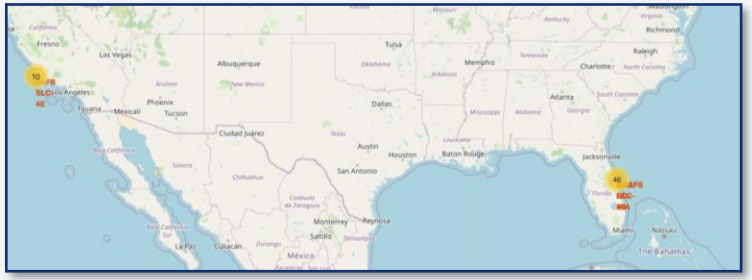
SQL queries were used to explore Falcon 9 launch records and mission outcomes

SQL Analysis Focus	Objective
Launch Sites	Retrieved records for specific launch sites and mission outcomes to compare launch activity
Orbit Types	Analyze launch success rates across orbit types and launch locations
Payload Mass	Investigate payload mass distributions and booster landing performance
Mission Outcomes	Examined mission history, flight frequency, and landing outcomes

GitHub ⬅ *Click here to view complete SQL analysis notebook on GitHub*

Build an Interactive Map with Folium

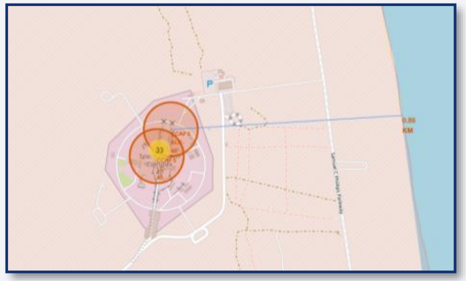
Folium maps enabled geographic analysis of launch sites, landing outcomes, and nearby infrastructure



Folium maps were used to visualize Falcon 9 launch site locations and landing outcomes geographically.

Launch sites are concentrated near coastal areas.

Color-coded markers highlighted successful and failed landings across launch facilities.

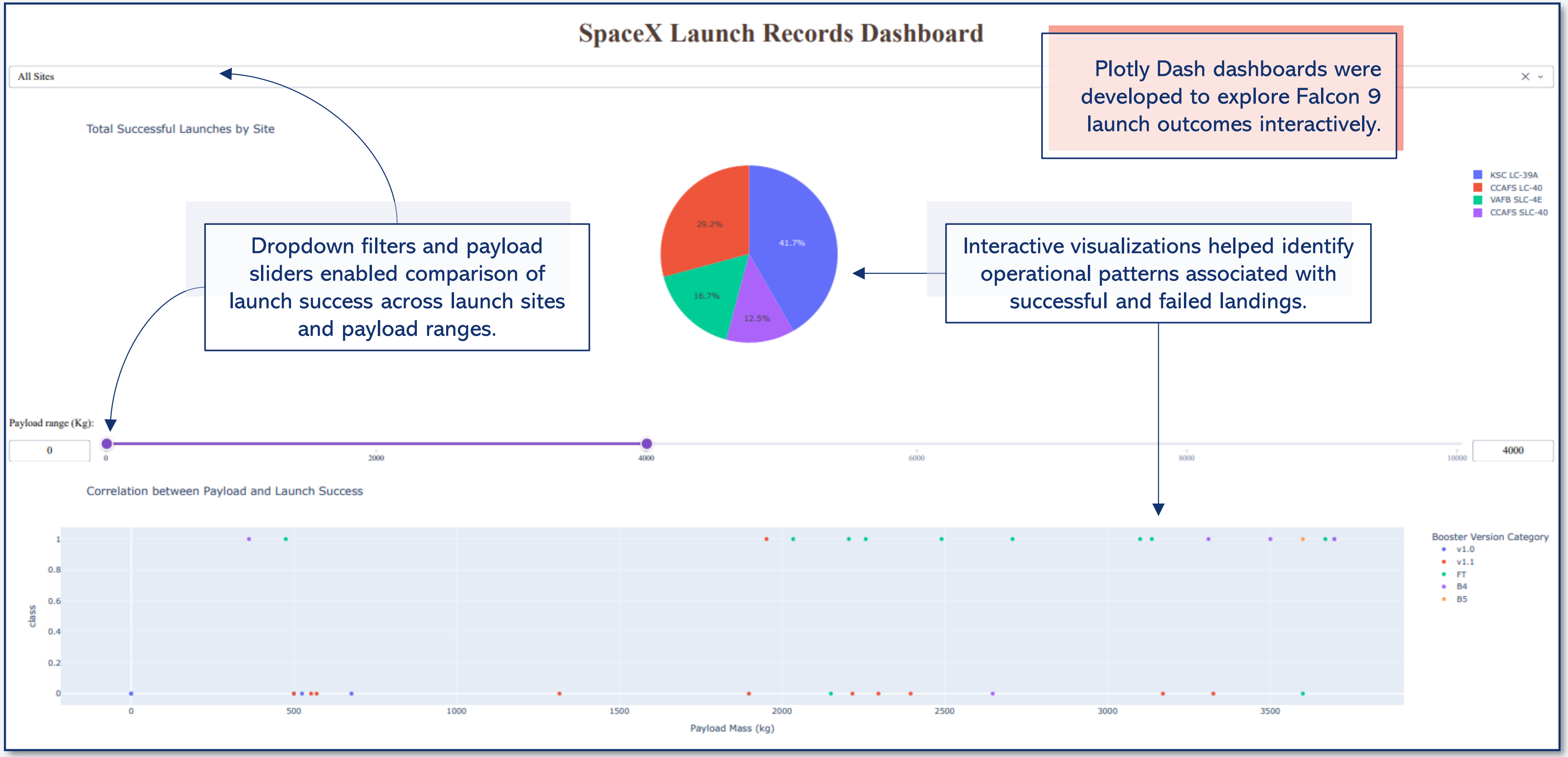


Distance calculations and geographic visualizations were used to analyze proximity between launch sites and nearby infrastructure.

GitHub ⬅ *Click here to view complete Folium notebook on GitHub*

Build a Dashboard with Plotly Dash

Interactive dashboards enabled dynamic exploration of Falcon 9 launch and landing patterns



Plotly Dash dashboards were developed to explore Falcon 9 launch outcomes interactively.

Dropdown filters and payload sliders enabled comparison of launch success across launch sites and payload ranges.

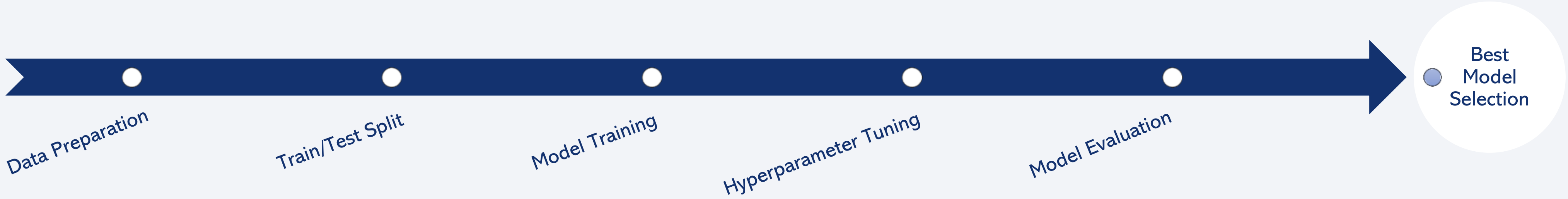
Interactive visualizations helped identify operational patterns associated with successful and failed landings.

GitHub ⬅ *Click here to view complete Plotly Dash application script on GitHub*

Predictive Analysis (Classification)

Machine learning classifiers were developed to predict Falcon 9 first stage landing success

1. Historical Falcon 9 launch variables were used to train classification models that predict first stage landing outcomes.
2. Logistic Regression, SVM, Decision Tree, and KNN classifiers were trained and compared using the prepared dataset.
3. GridSearchCV was used to optimize model hyperparameters and improve classification accuracy.
4. Model performance was evaluated using accuracy scores and confusion matrices.

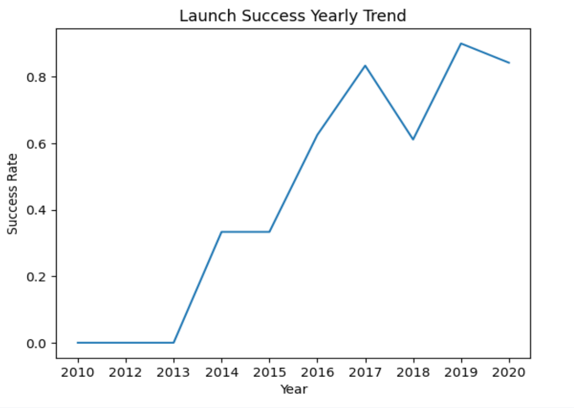


GitHub ⬅ *Click here to view complete predictive analysis notebook on GitHub*

Results Overview

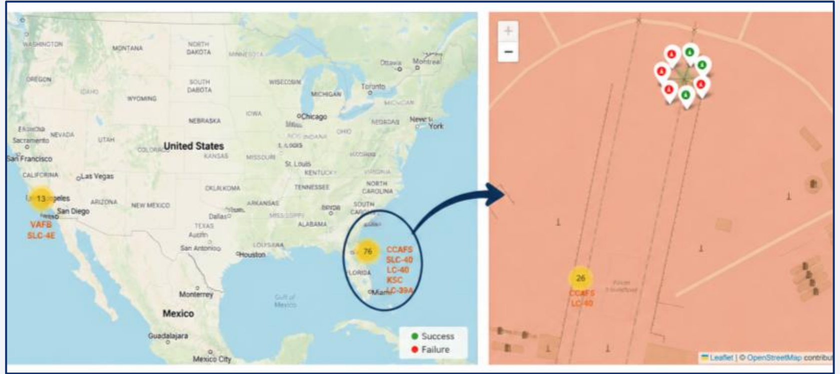
Exploratory analysis and classification models revealed patterns associated with Falcon 9 landing outcomes

Exploratory Analysis



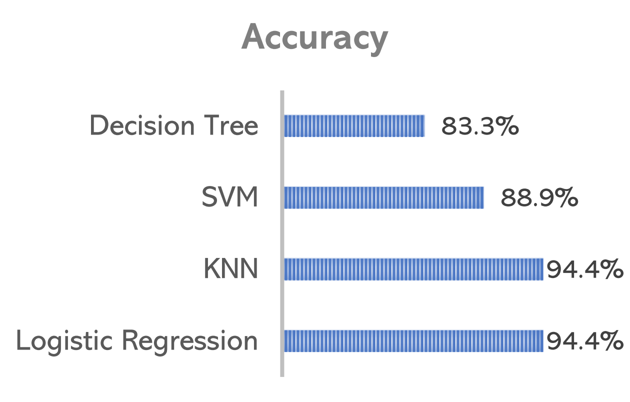
Landing success rates increased over time and varied across orbit types, payload ranges, and launch sites.

Interactive Analytics



Interactive dashboards and geographic visualizations supported comparative analysis of launch operations and landing outcomes.

Predictive Modeling



Classification models achieved high accuracy on the available Falcon 9 launch test dataset.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Landing success became more consistent over time across the major launch sites

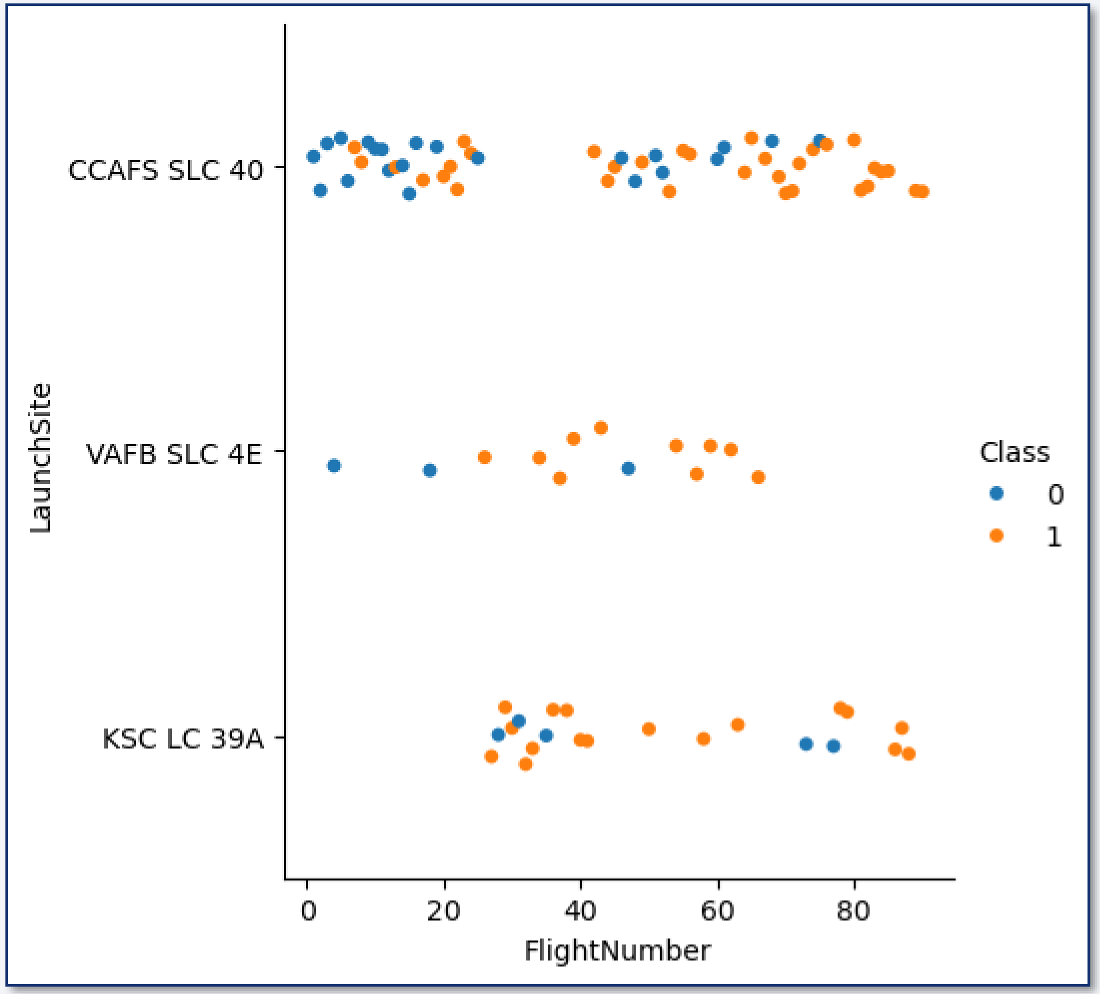


Figure: Flight Number vs. Launch Site colored by landing outcome

Flight number was used as a proxy for mission chronology to analyze launch performance over time.

Successful landings (Class = 1) became more frequent in later missions, particularly at CCAFS SLC 40.

The results suggest that Falcon 9 landing reliability improved as operational experience increased over time.

Class:
0 = Failure
1 = Success

Payload vs. Launch Site

Successful landings occurred across different payload ranges and launch sites

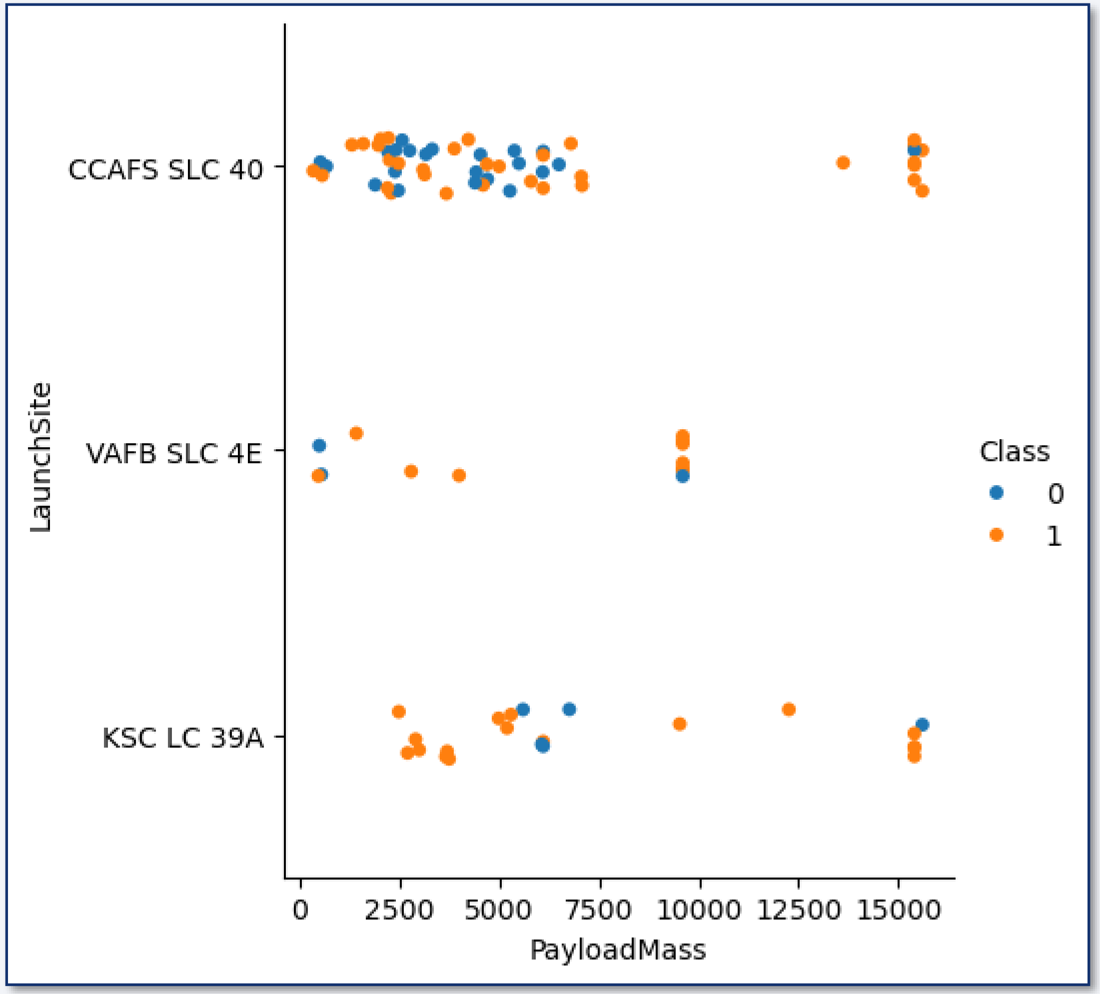


Figure: Payload Mass (kg) vs. Launch Site colored by landing outcome

Class:
0 = Failure
1 = Success

Payload mass was compared across launch sites to evaluate how mission capacity related to landing outcomes.

Successful landings were achieved across increasingly larger payload ranges, particularly at CCAFS SLC 40.

Higher payload missions increasingly achieved successful landings, suggesting improvements in Falcon 9 reusability and operational reliability.

Success Rate vs. Orbit Type

Landing success rates varied significantly across orbit types

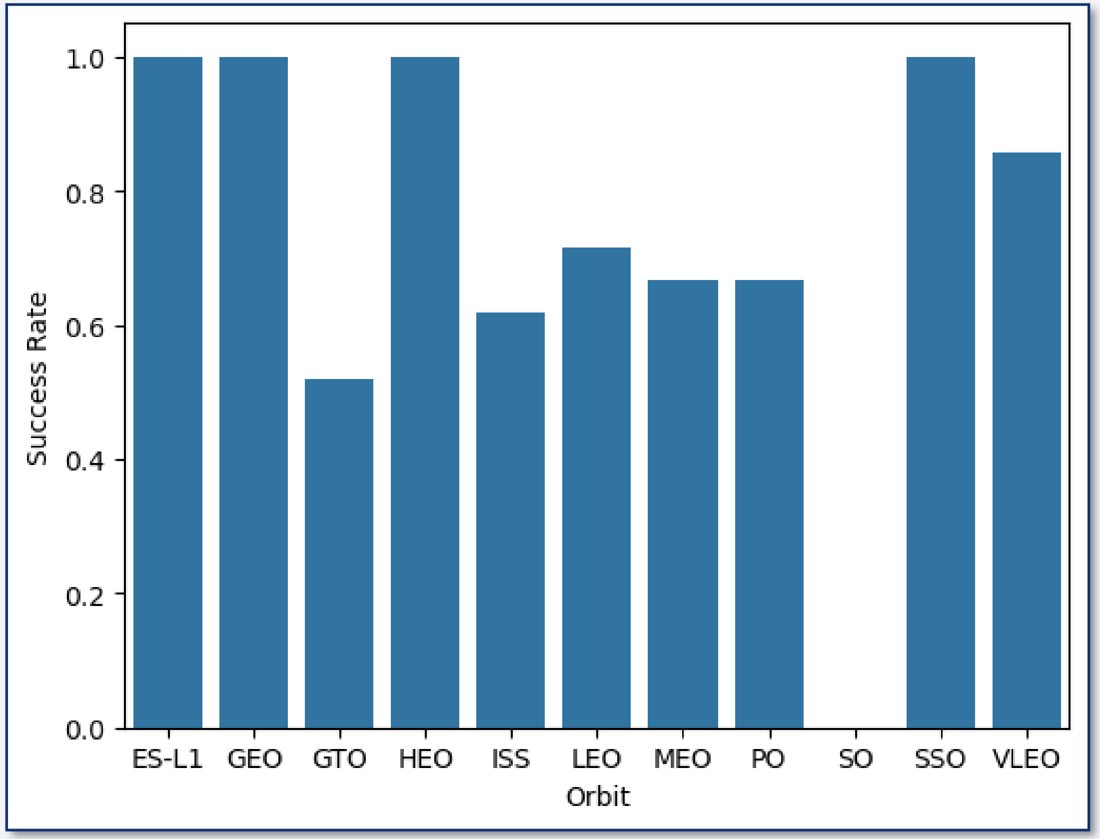


Figure: Falcon 9 landing success rate across orbit types

Some orbit categories contained limited observations, so individual success rates should be interpreted cautiously.

Orbit type was analyzed to evaluate how different mission profiles influenced Falcon 9 landing success rates.

Lower success rates in some orbit types may be associated with more demanding mission trajectories and landing conditions.

Landing success rates varied considerably across orbit categories, suggesting that some mission profiles are operationally more demanding than others.

Flight Number vs. Orbit Type

Successful landings became more frequent across multiple orbit types in later Falcon 9 missions

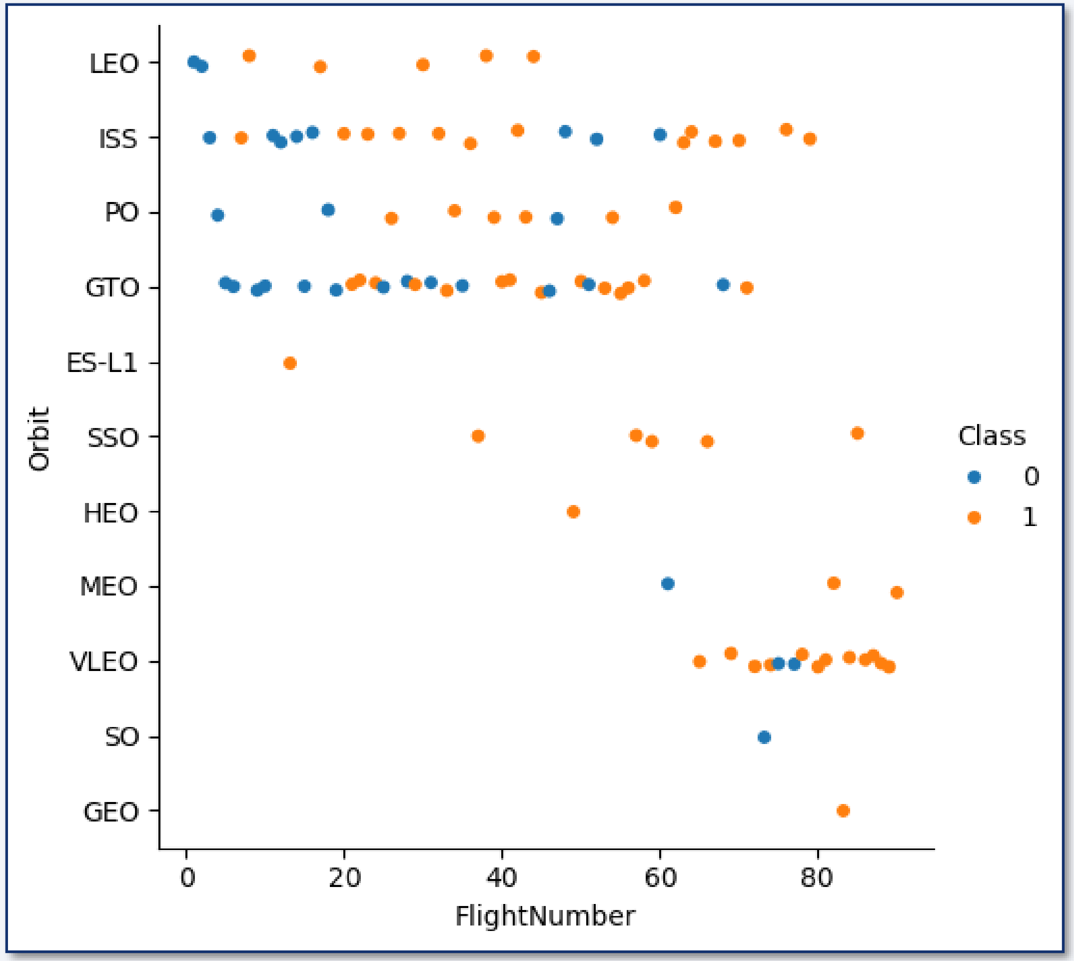


Figure: Flight Number vs. Orbit Type colored by landing outcome

Flight number was used to analyze how landing outcomes evolved over time across different orbit types.

Earlier missions showed a higher concentration of failed landings, particularly for GTO and ISS orbit missions.

Later Falcon 9 missions achieved more consistent landing success across multiple orbit categories.

Class:
0 = Failure
1 = Success

Payload vs. Orbit Type

Successful landings were achieved across multiple orbit types and payload capacities

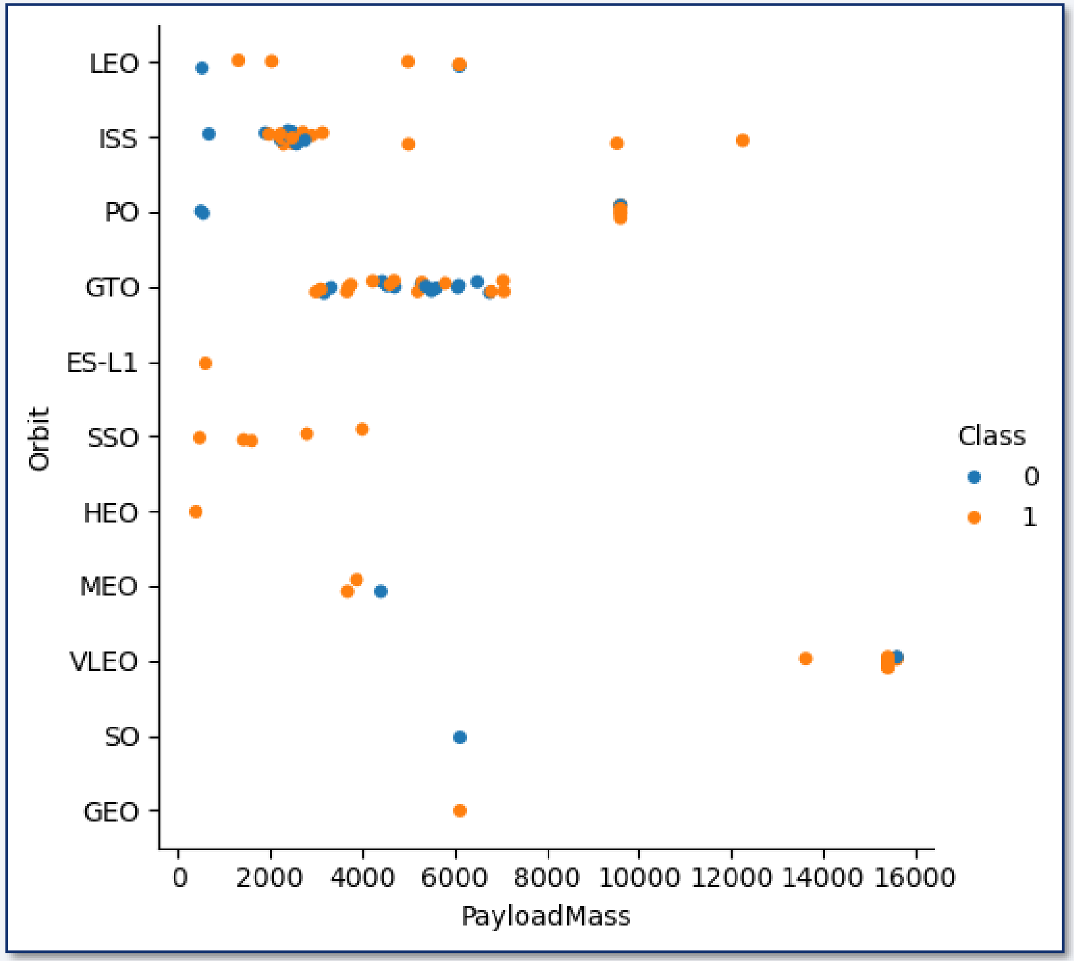


Figure: Payload Mass (kg) vs. Orbit Type colored by landing outcome

Payload mass was analyzed across orbit types to evaluate how mission profile related to landing outcomes.

Successful landings occurred across both low and high payload ranges, demonstrating Falcon 9 operational flexibility.

The results demonstrate Falcon 9 operational flexibility across diverse mission profiles and payload capacities.

Class:
0 = Failure
1 = Success

Launch Success Yearly Trend

Falcon 9 landing success rates improved significantly over time

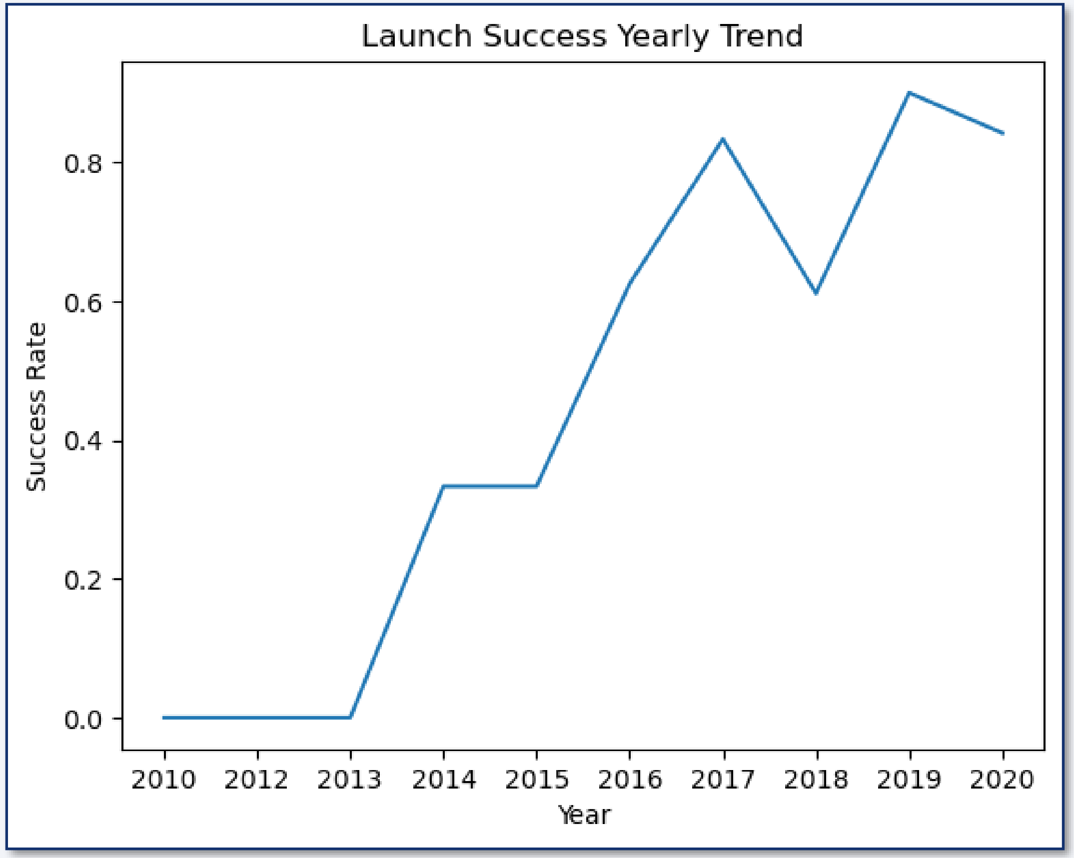


Figure: Average Falcon 9 landing success rate by mission year

Yearly average landing success rates were analyzed to evaluate long-term Falcon 9 operational performance.

Success rates increased substantially after 2013, reflecting the maturation of Falcon 9 reusable launch operations.

The upward trend demonstrates continuous improvements in landing reliability and mission execution over time.

All Launch Site Names

Multiple Falcon 9 launch sites were identified using SQL analysis.

Query

```
%%sql SELECT DISTINCT "Launch_Site"
FROM SPACEXTABLE
```

SQL queries were used to retrieve and explore launch site information from the mission database.

The DISTINCT statement was used to retrieve the unique Falcon 9 launch site names stored in the mission database.

Output

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

The query identified the four launch sites used across Falcon 9 missions.

First Successful Ground Landing Date

The analysis identified the first successful Falcon 9 landing on a ground pad

Query

```
%%sql SELECT MIN("Date") AS "Date"
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (ground pad)'
```

Query Result

Date
2015-12-22

SQL filtering was used to identify the first successful Falcon 9 ground pad landing.

The MIN function was used to retrieve the earliest date associated with a successful ground pad landing.

The first successful ground landing occurred on December 22nd, 2015 and marked an important milestone in Falcon 9 reusability operations.

Total Number of Successful and Failure Mission Outcomes

Mission outcome records were grouped to analyze Falcon 9 landing success patterns

Query

```
%%sql SELECT
    CASE
        WHEN "Landing_Outcome" LIKE 'Success%' THEN 'Success'
        WHEN "Landing_Outcome" LIKE 'Failure%' THEN 'Failure'
        ELSE 'Other'
    END AS Outcome,
    COUNT(*) AS "Number of Missions"
FROM SPACEXTABLE
GROUP BY Outcome
ORDER BY "Number of Missions" DESC
```

A CASE statement was used to categorize landing outcomes into Success, Failure, and Other mission outcomes.

The COUNT function aggregated the total number of missions for each outcome category.

Query Result

Outcome	Number of Missions
Success	61
Other	30
Failure	10

The analysis showed that successful mission outcomes occurred more frequently than failed missions.

Boosters Carried Maximum Payload

Multiple Falcon 9 boosters reached the highest payload capacity recorded in the dataset

Query

```
%%sql SELECT
    "Booster_Version",
    "PAYLOAD_MASS__KG_"
FROM SPACEXTABLE
WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE)
```

A subquery with the MAX function was used to identify the highest payload mass recorded in the dataset.

The query returned all booster versions associated with the maximum payload value of 15,600 kg.

Multiple Falcon 9 Block 5 boosters were associated with the highest payload value.

Query Result

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

Section 4

Launch Sites Proximities Analysis

Falcon 9 Launch Site Locations

Folium maps were used to visualize the geographic locations of Falcon 9 launch facilities

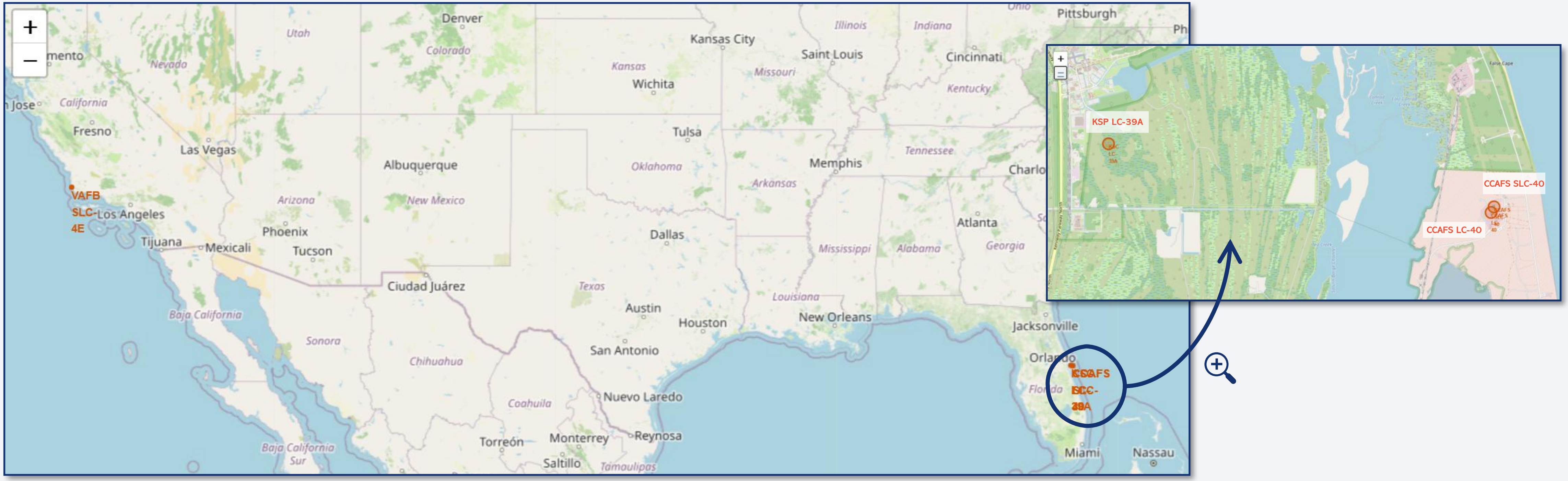


Figure: Geographic distribution of Falcon 9 launch sites visualized using Folium.

The map highlights launch facilities positioned near coastal regions, supporting safe launch trajectories and recovery operations.

The launch sites are in lower-latitude coastal regions, supporting more efficient launches.

Falcon 9 launch operations were distributed across four major launch sites in Florida and California.

Falcon 9 Landing Outcome Visualization

Color-coded Folium markers were used to compare successful and failed Falcon 9 landing attempts

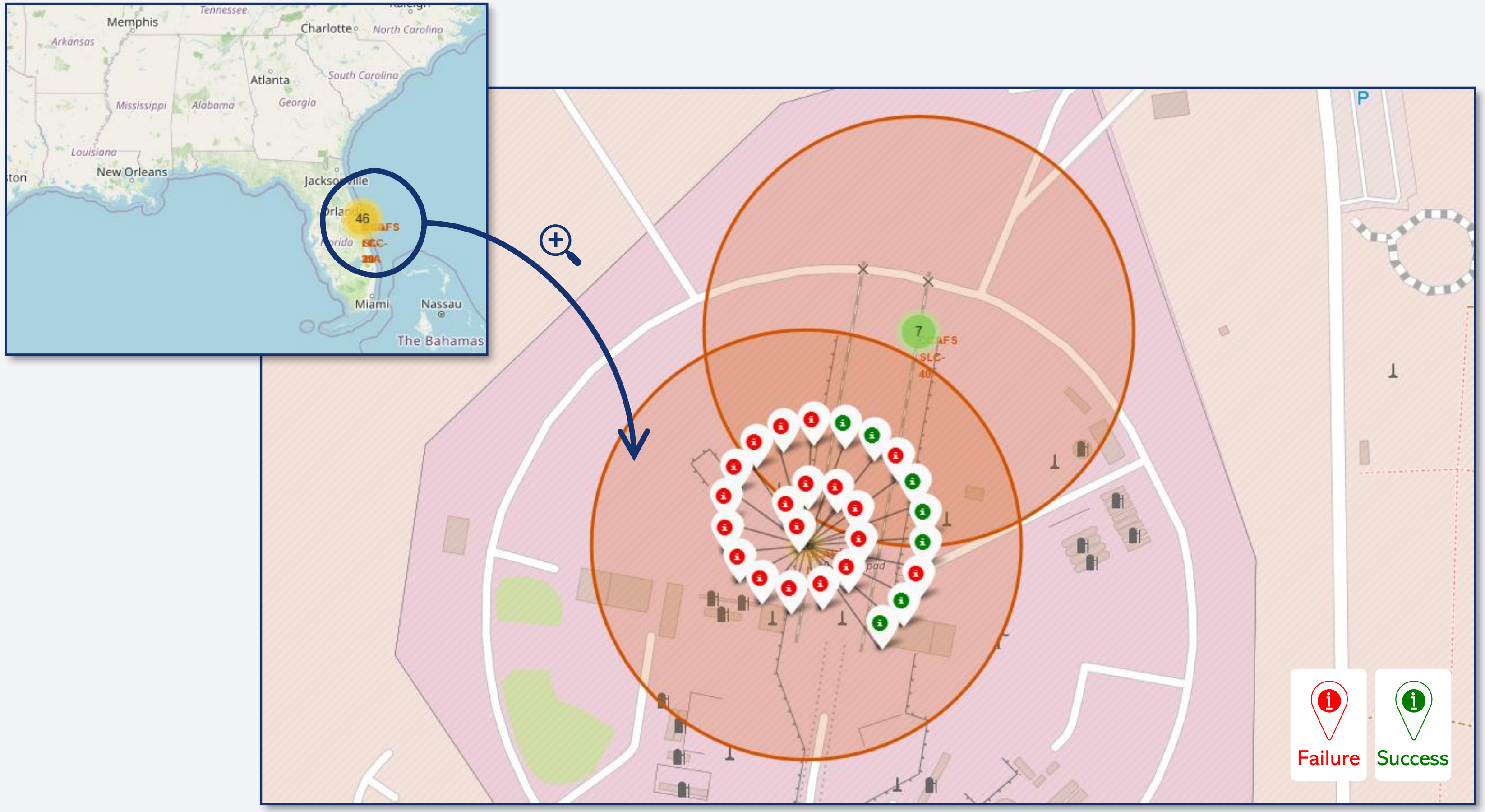


Figure: Folium map visualization of Falcon 9 landing outcomes at CCAFS LC-40 launch site.

Green and red markers were used to distinguish successful and failed Falcon 9 landing outcomes.

Marker clustering helped visualize the concentration of landing outcomes around the launch site.

Launch Site Proximity Analysis

Distance analysis of Falcon 9 launch site surroundings

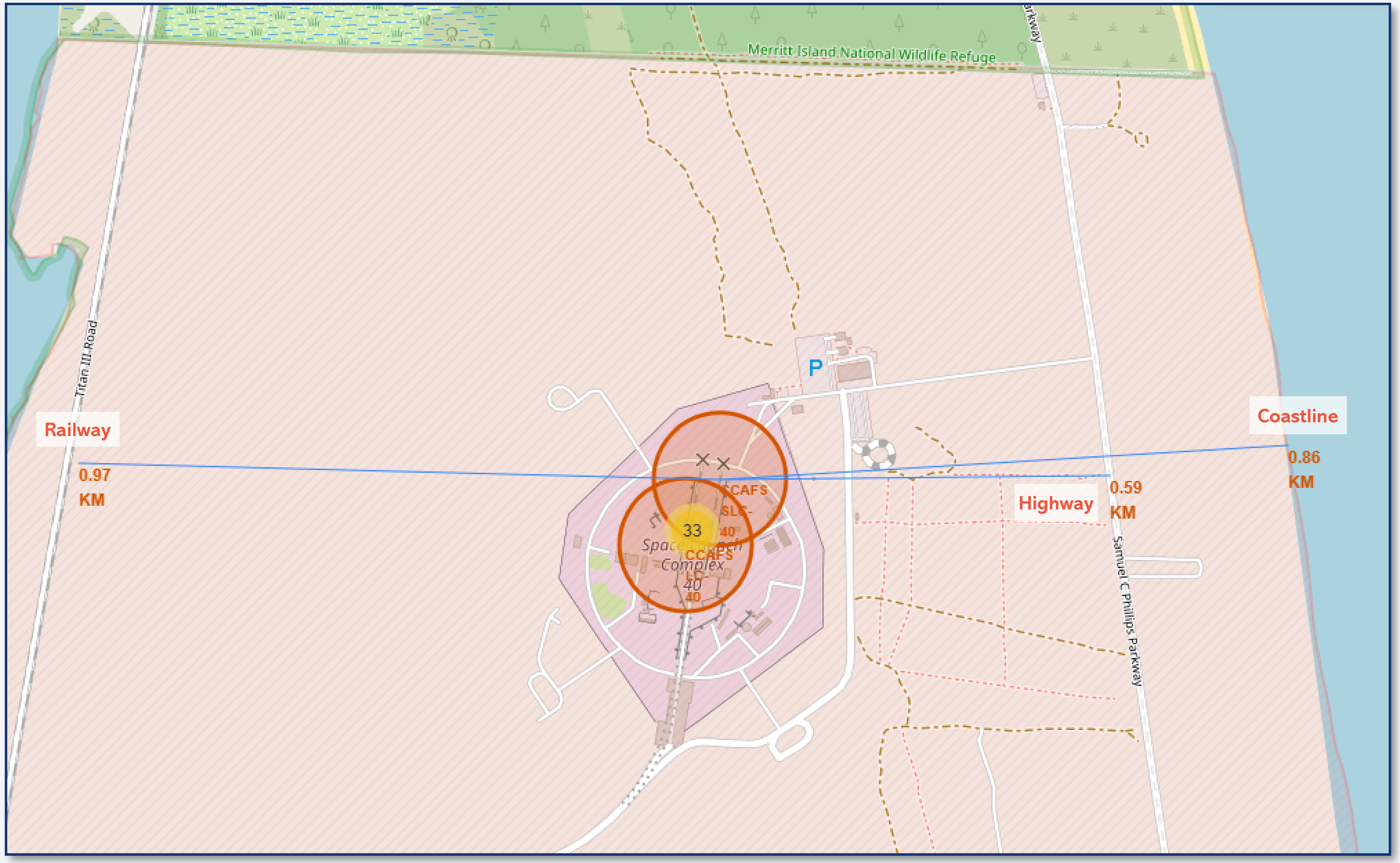


Figure: Distance analysis between CCAFS LC-40 and nearby infrastructure using Folium visualization.

Distance calculations and geographic visualizations were used to analyze the surroundings of the CCAFS LC-40 launch site.

The proximity to highways, railways, and coastline highlights the strategic geographic positioning of the launch facility.

The analysis showed that CCAFS LC-40 is positioned near transportation routes and coastal areas that support launch and recovery operations.

Section 5

Build a Dashboard with Plotly Dash

Launch Success Distribution by Site

The pie chart shows the distribution of successful Falcon 9 launches across launch sites

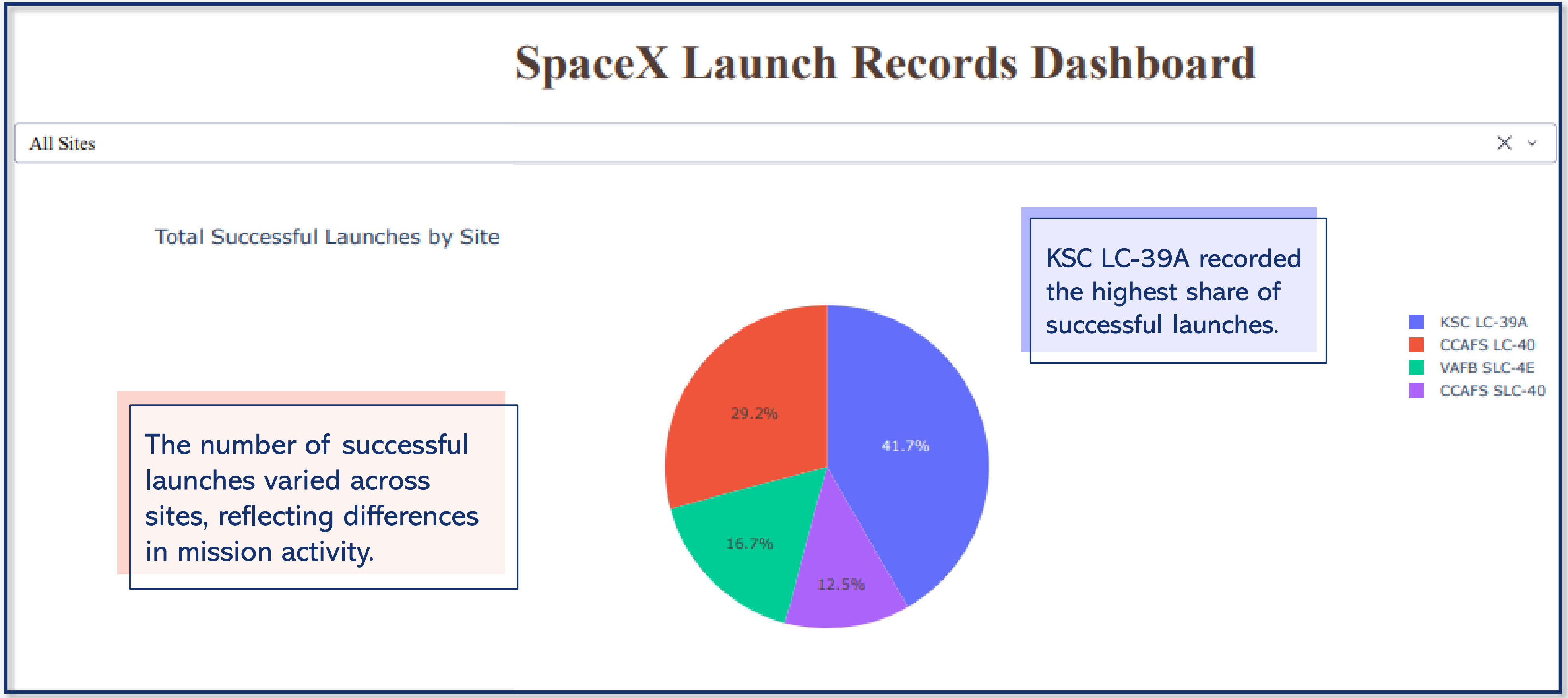
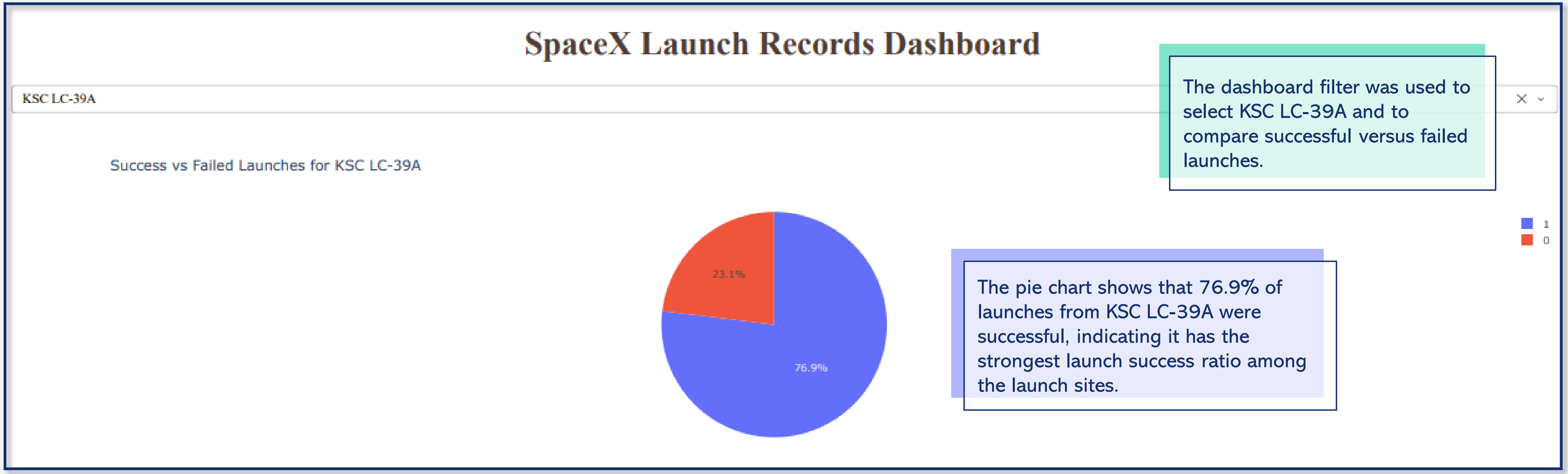


Figure: Pie chart showing the distribution of successful Falcon 9 launches across all launch sites.

Highest Launch Success Ratio — KSC LC-39A

KSC LC-39A recorded the highest launch success ratio in the dashboard analysis



The dashboard filter was used to select KSC LC-39A and to compare successful versus failed launches.

The pie chart shows that 76.9% of launches from KSC LC-39A were successful, indicating it has the strongest launch success ratio among the launch sites.

Figure: Success vs. failed Falcon 9 launches for KSC LC-39A using the Plotly Dash dashboard.

Payload Mass vs. Launch Outcome Analysis

Dashboard filters were used to compare launch outcomes across payload ranges.

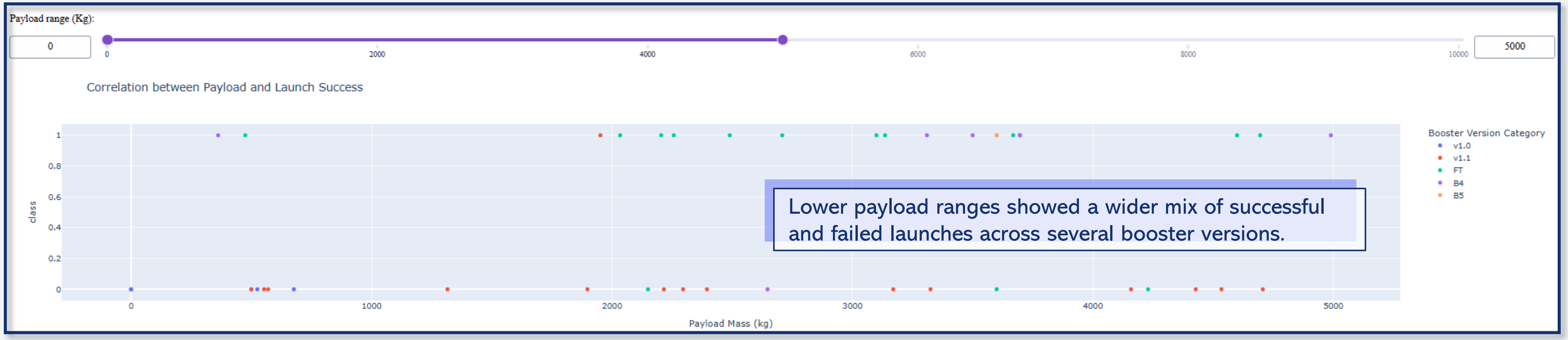


Figure 1: Launch outcomes for payloads between 0 and 5000 kg.

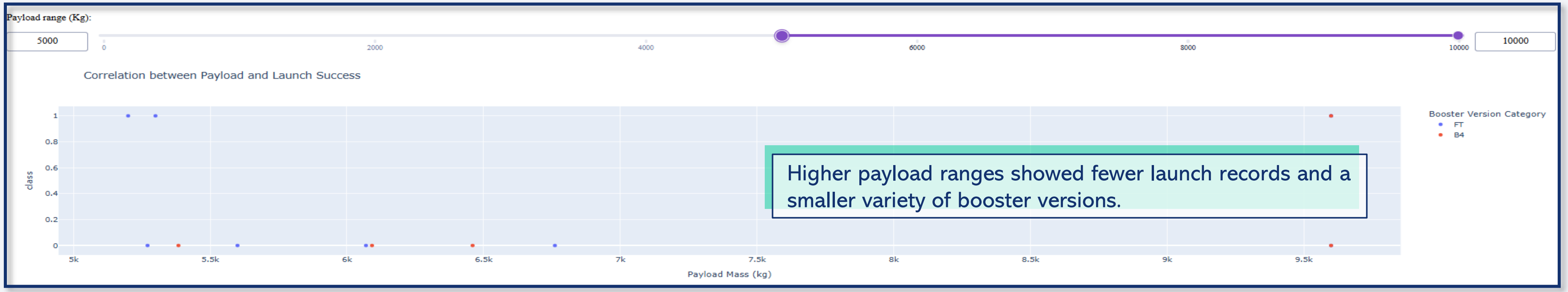


Figure 2: Launch outcomes for payloads between 5000 and 10000 kg.

Successful launches were observed across both payload ranges.

Section 6

Predictive Analysis (Classification)

Classification Model Accuracy Comparison

Classification model accuracy scores were compared across the evaluated algorithms

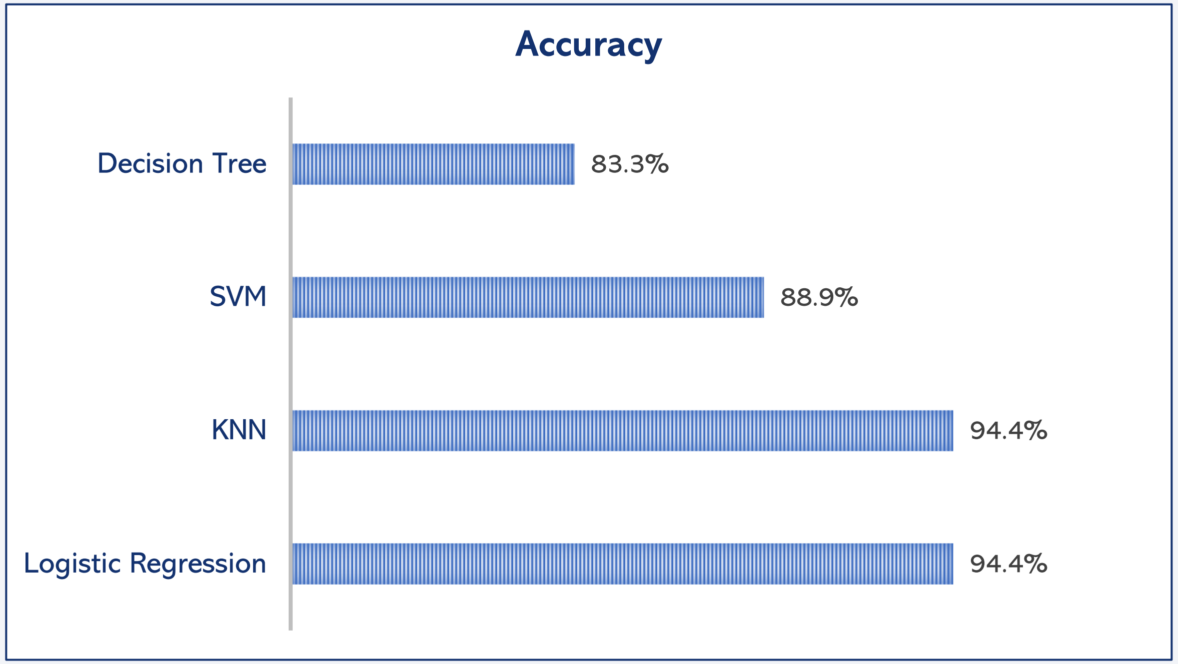


Figure: Accuracy comparison across the evaluated classification models.

KNN and Logistic Regression achieved the highest classification accuracy on the test dataset at 94.4%.

SVM also performed well, while Decision Tree achieved the lowest accuracy among the evaluated models.

The results suggest that historical Falcon 9 launch variables contain predictive patterns associated with landing outcomes.

Confusion Matrix - Logistic Regression

Confusion matrix analysis was used to evaluate classification performance on the test dataset

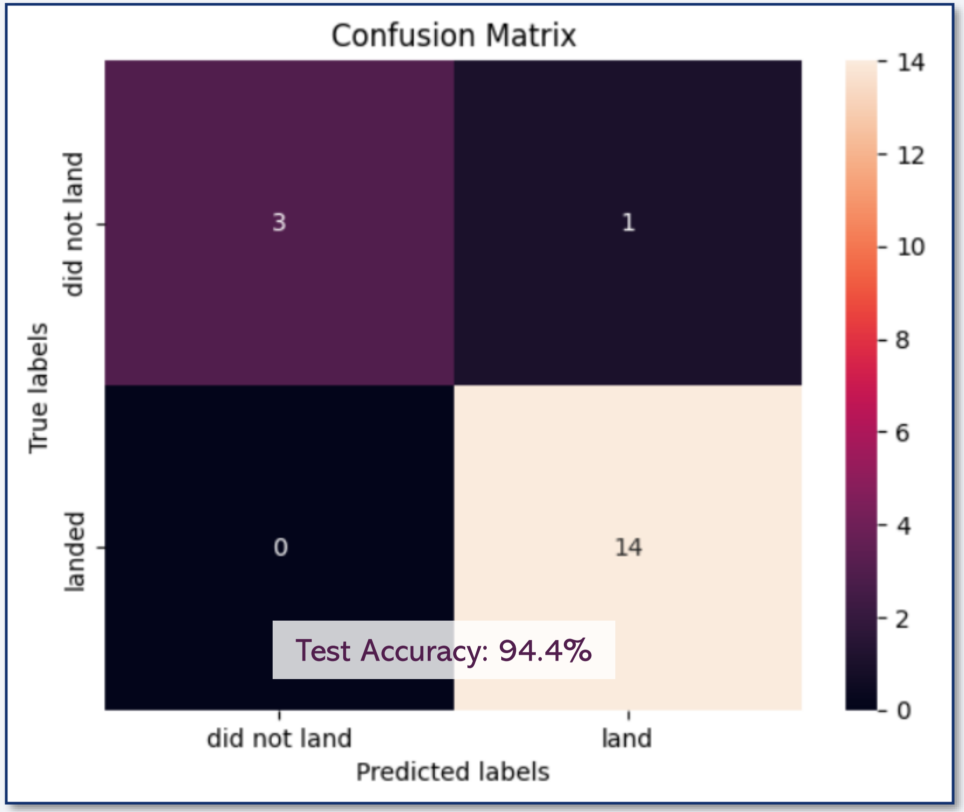


Figure: Confusion matrix for the Logistic Regression classification model.

The Logistic Regression model correctly classified most launch outcomes in the test dataset.

Only one failed landing was incorrectly predicted as a successful landing.

KNN achieved the same accuracy score and identical confusion matrix results as Logistic Regression.

Conclusions

1. Exploratory data analysis identified relationships between launch site, orbit type, payload mass, and Falcon 9 landing outcomes.
2. Landing success rates improved substantially over time, reflecting the maturation of Falcon 9 reusable launch operations.
3. Interactive SQL analysis, Folium maps, and Plotly Dash dashboards supported exploration of operational and geographic launch patterns.
4. KSC LC-39A recorded the highest launch success ratio among the analyzed launch sites.
5. Classification models achieved high accuracy when predicting Falcon 9 first stage landing outcomes using historical launch variables.
6. Logistic Regression and KNN achieved the best overall classification performance on the test dataset.

Overall, the project demonstrates how data science techniques can be applied to analyze and predict reusable rocket landing performance.

Appendix

Contents

Appendix - Dataset Sample

Sample of the processed Falcon 9 launch dataset used for exploratory analysis and predictive modeling.

FlightNumber	BoosterVersion	PayloadMass	Orbit	LaunchSite	Class
1	Falcon 9	6104.9594	LEO	CCAFS SLC 40	0
2	Falcon 9	525	LEO	CCAFS SLC 40	0
3	Falcon 9	677	ISS	CCAFS SLC 40	0
4	Falcon 9	500	PO	VAFB SLC 4E	0
5	Falcon 9	3170	GTO	CCAFS SLC 40	0
6	Falcon 9	3325	GTO	CCAFS SLC 40	0
7	Falcon 9	2296	ISS	CCAFS SLC 40	1
8	Falcon 9	1316	LEO	CCAFS SLC 40	1
9	Falcon 9	4535	GTO	CCAFS SLC 40	0
10	Falcon 9	4428	GTO	CCAFS SLC 40	0
11	Falcon 9	2216	ISS	CCAFS SLC 40	0
12	Falcon 9	2395	ISS	CCAFS SLC 40	0

Filtering Launch Site with SQL

SQL filtering was used to retrieve launch records by launch site pattern

Query

```
%%sql SELECT *
FROM SPACEXTABLE
WHERE "Launch_Site" LIKE 'CCA%'
LIMIT 5
```

The LIKE operator was used to filter records where the launch site name starts with ‘CCA’.

Query result

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The query returned launch records associated with Cape Canaveral launch facilities.

SQL filtering techniques supported targeted exploration of Falcon 9 mission data.

Total Payload Mass

SQL aggregation functions were used to calculate the total payload mass delivered for NASA CRS missions

Query

```
%%sql SELECT Customer, SUM("PAYLOAD_MASS__KG_") AS "TotalPayloadMass_Kg"
FROM SPACEXTABLE
WHERE "Customer" = 'NASA (CRS)'
GROUP BY "Customer"
```

Query result

Customer	TotalPayloadMass_Kg
NASA (CRS)	45596

The query aggregated payload values across multiple launches associated with NASA cargo resupply operations.

The SUM function was used to calculate the cumulative payload mass for NASA CRS missions.

The analysis showed that Falcon 9 boosters delivered a total payload mass of 45,596 kg for NASA CRS missions which highlights the significant payload capacity used in NASA CRS missions.

Average Payload Mass by F9 v1.1

Average payload mass was analyzed for the Falcon 9 v1.1 booster version

Query

```
%%sql SELECT "Booster_Version", AVG("PAYLOAD_MASS__KG_") AS "AVGPayloadMass_Kg"
FROM SPACEXTABLE
WHERE "Booster_Version" = 'F9 v1.1'
GROUP BY "Booster_Version"
```

The query grouped payload records by booster version to analyze average payload capacity.

The AVG function was used to calculate the average payload mass carried by the Falcon 9 v1.1 booster.

Query Result

Booster_Version	AVGPayloadMass_Kg
F9 v1.1	2928.4

Falcon 9 v1.1 missions carried an average payload mass of nearly 3000 kg.

Successful Drone Ship Landing with Payload between 4000 and 6000

Successful drone ship landings were analyzed for payloads between 4,000 kg and 6,000 kg

Query

```
%%sql SELECT "Booster_Version", "PAYLOAD_MASS__KG_"
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (drone ship)'
AND "PAYLOAD_MASS__KG_" > 4000
AND "PAYLOAD_MASS__KG_" < 6000
```

The query filtered missions with successful drone ship landings and payload masses between 4,000 kg and 6,000 kg.

Conditional SQL operators were used to isolate missions within a specific payload interval.

Query Result

Booster_Version	PAYLOAD_MASS__KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

The results identified several Falcon 9 boosters that successfully landed on drone ships while carrying medium-range payload masses between 4,000 kg and 6,000 kg.

2015 Launch Records

Only a small number of drone ship landing failures were recorded during Falcon 9 missions in 2015

Query

```
%%sql SELECT
    SUBSTR("Date", 6, 2) AS MONTH,
    "Landing_Outcome",
    "Booster_Version",
    "Launch_Site"
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Failure (drone ship)'
AND SUBSTR("Date", 0, 5) = '2015'
```

The query filtered launch records for drone ship landing failures that occurred during 2015.

The SUBSTR function was used to extract the year and month values from the mission date field.

Result

MONTH	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

The analysis identified two drone ship landing failures recorded during Falcon 9 missions in 2015, both launched from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

SQL aggregation was used to rank Falcon 9 landing outcomes between 2010 and 2017

Query

```
%%sql SELECT
    "Landing_Outcome",
    COUNT("Landing_Outcome") AS TotalOutcomes
FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY "TotalOutcomes" DESC
```

The query grouped and ranked landing outcomes recorded between June 2010 and March 2017.

COUNT and GROUP BY were used to calculate the frequency of each landing outcome category.

Query Result

Landing_Outcome	TotalOutcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

"No attempt" was the most common outcome during the early Falcon 9 missions, while successful drone ship and ground pad landings occurred at similar frequencies.

Appendix - Model Feature Table

Classification Features

Flight Number

Payload Mass

Orbit Type

Launch Site

Booster Version Category

Landing Outcome (Target Variable)

Appendix - Data Preparation Code

Example Python code snippets used for feature engineering, train/test splitting, and classification model training

```
URL = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_2.csv"
df=pd.read_csv(URL)
features = df[['FlightNumber', 'PayloadMass', 'Orbit', 'LaunchSite', 'Flights', 'GridFins',
               'Reused', 'Legs', 'LandingPad', 'Block', 'ReusedCount', 'Serial']]
features_one_hot = pd.get_dummies(
    features,
    columns=['Orbit', 'LaunchSite', 'LandingPad', 'Serial']
)
features_one_hot = features_one_hot.astype('float64')
features_one_hot.to_csv('dataset_part\_3.csv', index=False)
```

Feature Engineering

```
URL2 = 'https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_3.csv'
X = pd.read_csv(URL2)
Y = data['Class'].to_numpy()
transform = preprocessing.StandardScaler()
X = transform.fit_transform(X)
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
```

Train/Test Split

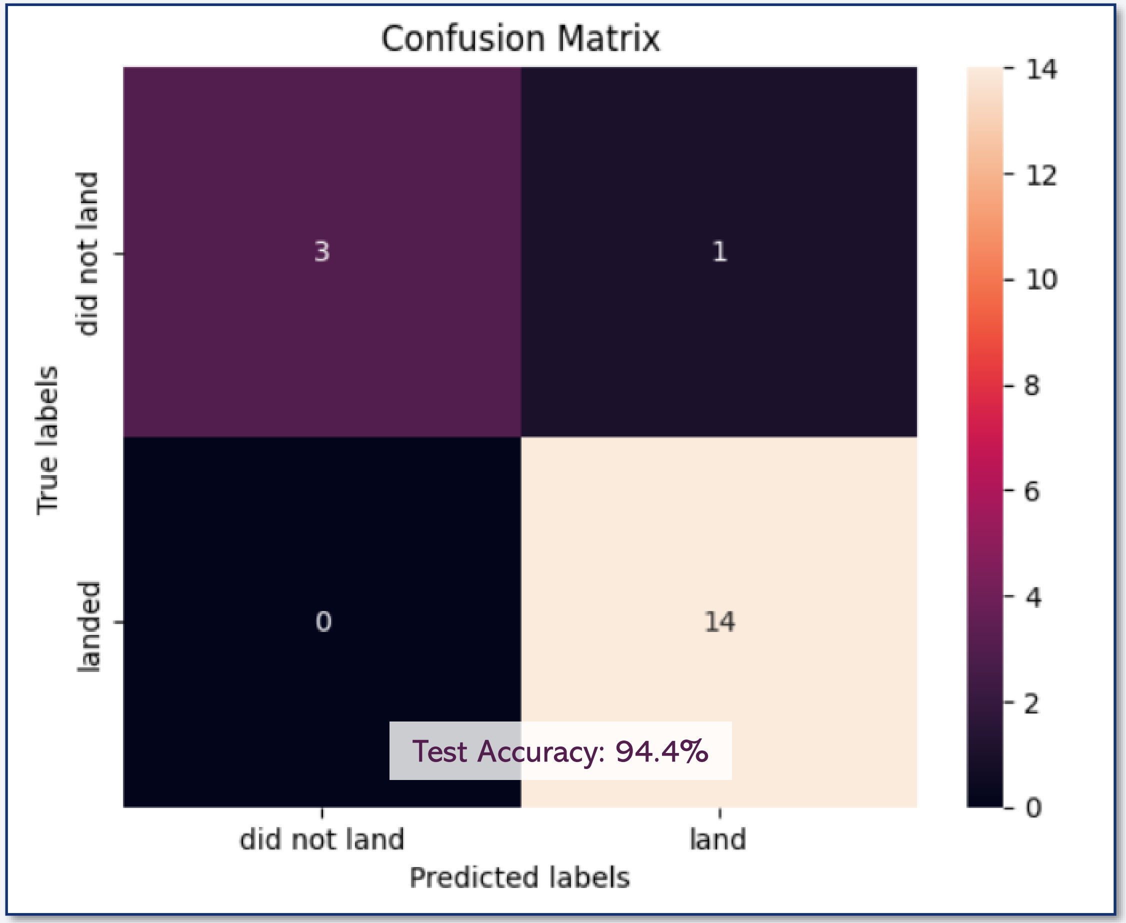
```
parameters ={'C':[0.01,0.1,1],
             'penalty':['l2'],
             'solver':['lbfgs']}
lr=LogisticRegression()

logreg_cv = GridSearchCV(
    estimator=lr,
    param_grid=parameters,
    cv=10
)

logreg_cv.fit(X_train, Y_train)
```

Model fitting

(Logistic Regression)

Appendix - KNN Confusion Matrix

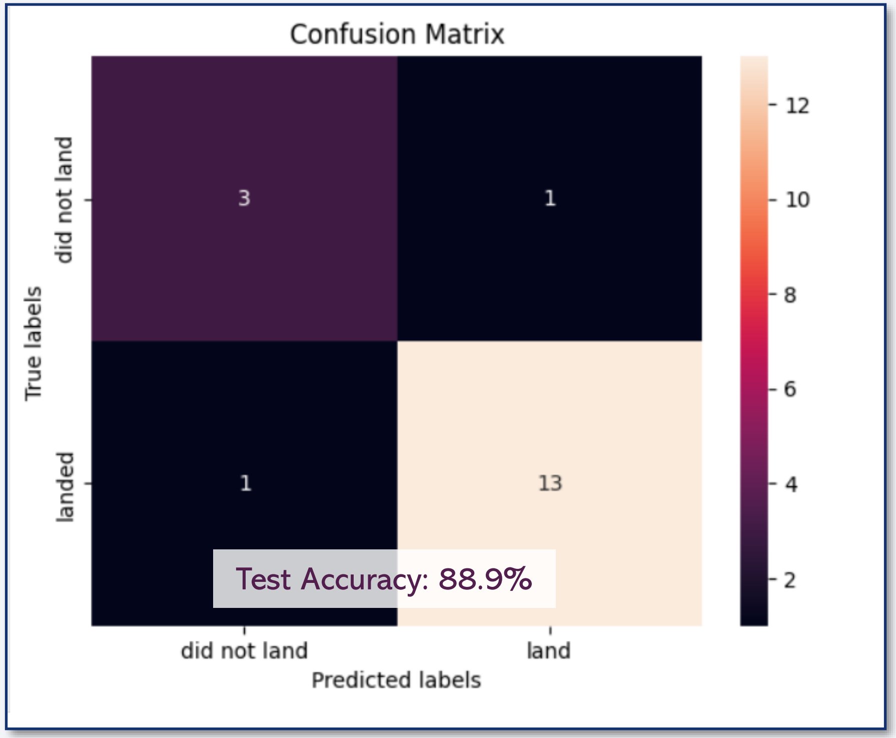
```
yhat = knn_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



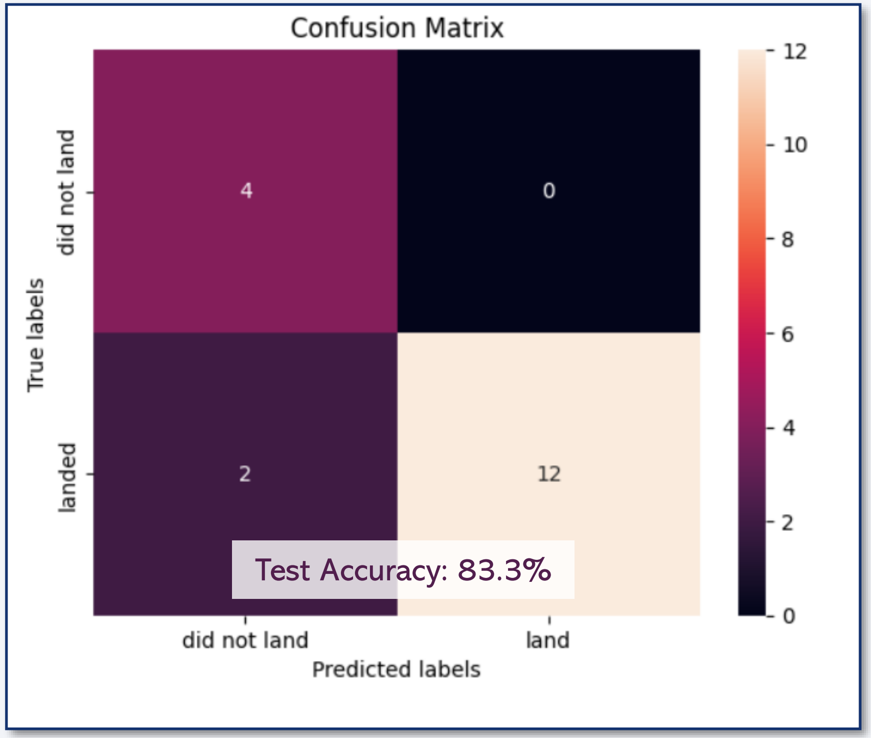
KNN achieved the same accuracy score and identical confusion matrix results as Logistic Regression.

Appendix – SVM/Decision Tree Confusion Matrix

```
yhat=svm_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```




```
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



SVM and Decision Tree confusion matrices are presented for comparison with the best-performing classification models.

Appendix – GitHub Repository link

GitHub

Click here to view complete GitHub Repository

Thank you!